

Tools — Git

Local and remote git operations. Remote operations (`git_push`, destructive resets) follow the Version Control Safety rules: never push unless the user explicitly asks; never rewind files with `git_checkout`.

Tool	Description
<code>git_add</code>	Stage files for commit.
<code>git_branch</code>	List local and remote branches.
<code>git_checkout</code>	Switch branch or restore files.
<code>git_cherry_pick</code>	Apply changes from existing commits.
<code>git_clone</code>	Clone a repository.
<code>git_commit</code>	Create a commit from staged changes.
<code>git_diff</code>	Show working tree, staged, or commit diff.
<code>git_fetch</code>	Fetch from remote without merging.
<code>git_log</code>	Show commit history.
<code>git_merge</code>	Merge another branch.
<code>git_pull</code>	Fetch and integrate from remote.
<code>git_push</code>	Push branches and tags to remote.
<code>git_remote</code>	List, add, remove, update remotes.
<code>git_reset</code>	Unstage or reset to a commit.
<code>git_show</code>	Show commit, tag, or object details.
<code>git_stash</code>	Stash and unstash changes.
<code>git_status</code>	Show working tree status.
<code>git_tag</code>	List, create, show, or delete tags.

`git_add`

Stage files for the next commit. At least one of `paths`, `all`, or `update` must be provided.

Arguments

Argument	Type	Required	Description	Typical value
<code>paths</code>	array	one of three	Specific files or directories to stage	<code>["src/main.rs"]</code>
<code>all</code>	boolean	one of three	Stage all changes (<code>git add -A</code>)	<code>true</code>
<code>update</code>	boolean	one of three	Stage changes to tracked files only (<code>git add -u</code>)	<code>true</code>

`git_branch`

List local and remote branches, including the checked-out branch.

Arguments

Argument	Type	Required	Description	Typical value
all	boolean	no	Include remote-tracking branches	true
format	enum	no	short (terse names) or verbose (with upstream tracking info)	"short"

git_checkout

Switch branches, or restore working-tree files. Provide exactly one mode. **Do not use to rewind files** — it risks losing work.

Arguments

Argument	Type	Required	Description
branch	string	branch mode	Branch to switch to
create_branch	boolean	no	Create and switch (git checkout -b)
paths	array	restore mode	Files/directories to restore
source	string	no	Ref to restore from (default HEAD; only with paths)

git_cherry_pick

Apply changes from existing commits onto the current branch.

Arguments

Argument	Type	Required	Description
commits	array	yes	Commit hashes or refs to cherry-pick (first must be valid)
no_commit	boolean	no	Apply to working tree/index without committing (-n)

git_clone

Clone a repository into a new directory inside the working directory.

Arguments

Argument	Type	Required	Description	Typical value
url	string	yes	Repository URL	"https://github.com/user/repo.git"
directory	string	no	Target folder name (default: inferred from URL)	—
branch	string	no	Branch to check out (--branch)	"main"
depth	integer	no	Shallow clone depth (--depth)	1
bare	boolean	no	Create a bare repository	false

git_commit

Create a commit from staged changes.

Arguments

Argument	Type	Required	Description	Typical value
message	string	yes	Commit message	"Fix config loader panic"
all	boolean	no	Also stage modified tracked files (-a)	false
amend	boolean	no	Amend the previous commit	false
no_verify	boolean	no	Bypass pre-commit hooks	false

git_diff

Show differences for the working tree, the staged index, or a commit.

Arguments

Argument	Type	Required	Description	Typical value
target	string	no	working (default), staged/cached, or a commit ref	"staged"

Argument	Type	Required	Description	Typical value
path	string	no	Limit to one file or directory	"src/cli.rs"
stat	boolean	no	Compact change summary only	true

git_fetch

Fetch updates from a remote without merging.

Arguments

Argument	Type	Required	Description
remote	string	no	Remote name (default origin)
branch	string	no	Specific branch or ref
prune	boolean	no	Remove stale remote-tracking branches
all	boolean	no	Fetch every remote (ignores remote/branch)

git_log

Show commit history for the current branch or a ref.

Arguments

Argument	Type	Required	Description	Typical value
limit	integer	no	Max commits	20
branch	string	no	Ref to log (default: current branch/HEAD)	"main"
oneline	boolean	no	Compact one-line format	true
author	string	no	Filter by author name or email	—
since	string	no	Only newer commits	"2024-01-01", "1.week"

git_merge

Merge another branch into the current branch.

Arguments

Argument	Type	Required	Description
branch	string	yes	Branch to merge in
message	string	no	Custom merge commit message
no_ff	boolean	no	Always create a merge commit
ff_only	boolean	no	Abort unless fast-forward (mutually exclusive with no_ff)
squash	boolean	no	Squash incoming changes into one commit

git_pull

Fetch and integrate changes into the current branch.

Arguments

Argument	Type	Required	Description
remote	string	no	Remote name (default origin)
branch	string	no	Remote branch to pull (default: tracking branch)
rebase	boolean	no	Rebase instead of merge

git_push

Push local branches and tags to a remote. **Only call with explicit user instruction.**

Arguments

Argument	Type	Required	Description
remote	string	no	Remote name (default origin)
branch	string	no	Branch to push (default: current)
force	boolean	no	Force with lease (destructive; requires explicit confirmation)
tags	boolean	no	Push all tags

git_remote

List, add, remove, or set URLs for remotes.

Arguments

Argument	Type	Required	Description
action	enum	no	list (default), add, remove, set-url
name	string	yes for non-list	Remote name
url	string	yes for add/set-url	Remote URL

git_reset

Unstage files or reset the branch to a commit. Choose one mode. **hard discards all local working-tree changes.**

Arguments

Argument	Type	Required	Description	Typical value
paths	array	unstage mode	Files to unstage	["src/x.rs"]
target	string	reset mode	Commit hash, tag, or ref (default HEAD)	"HEAD~1"
mode	enum	reset mode	mixed (default, keeps changes unstaged), soft (keeps staged), hard, keep	"mixed"

git_show

Show details of a commit, tag, or other git object.

Arguments

Argument	Type	Required	Description	Typical value
ref	string	no	Commit hash, tag, or ref (default HEAD)	"abc123"
stat	boolean	no	Include file change statistics (default true)	true

git_stash

Stash and unstash working-tree changes.

Arguments

Argument	Type	Required	Description
action	enum	no	push (default), pop, apply, drop, list, clear
message	string	no	Message for push
index	integer	no	Stash index for pop/apply/drop (default 0)

git_status

Show working-tree status: modified, staged, untracked, conflicted files.

Arguments

Argument	Type	Required	Description
short	boolean	no	Compact one-line-per-file format
branch	boolean	no	Include current branch name (default true)

git_tag

List, show, create, or delete tags. Annotated tags require both name and message. **Do not tag releases unless explicitly asked.**

Arguments

Argument	Type	Required	Description
action	enum	no	list (default), show, create, delete
name	string	yes for show/create/delete	Tag name
message	string	annotated create	Tag message
ref	string	no	Target commit or ref (default HEAD)

Example:

```
git_status
git_add paths=["src/main.rs"]
git_commit message="Add greet subcommand"
```